Pre-Lab CKD Early Detection Using Random Forest: Evidence from NHANES and a Feature-Tier Ablation of the UCI CKD Dataset

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Course and section: _____

***Abstract* - Chronic kidney disease (CKD) is often found late because it has no symptoms until kidney function is badly damaged. Machine learning studies on the UCI CKD dataset report close to 100% accuracy, but they use serum creatinine, urea and urine albumin, which are the measurements that define CKD. I asked whether CKD can be screened for using only information available before any laboratory test. Using NHANES 2017-March 2020, I built a cohort of 7,363 US adults never told they had kidney disease (16.5% with CKD by eGFR or albuminuria) and compared Logistic Regression, Random Forest and Gradient Boosting on 13 pre-laboratory features, with F2 as the primary metric and identical folds, preprocessing and threshold tuning. The three models were nearly identical (cross-validated F2 0.578 to 0.580). Random Forest was selected by a pre-registered rule and reached F2 0.586, recall 0.864 and ROC-AUC 0.773 on the held-out test set, which is close to a simple risk-factor rule (F2 0.583). On UCI, the same algorithms reached 99% accuracy with all features, but with only pre-laboratory features their F2 was no better than referring everyone. Pre-laboratory screening is feasible but limited; the high accuracy in the literature comes from features that are part of the diagnosis.**

***Index Terms* - chronic kidney disease, screening, machine learning, data leakage, NHANES**

I. INTRODUCTION

Chronic kidney disease affected an estimated 697.5 million people worldwide in 2017 and is a growing cause of death [1]. It is defined by a reduced glomerular filtration rate (GFR below 60 mL/min/1.73m2) or markers of kidney damage such as albuminuria, present for more than three months [2]. Early CKD has no symptoms, so it is only found if someone is tested. In the Philippines, as in many low- and middle-income settings, many patients are first diagnosed when they already need dialysis. A screening tool that a rural health unit could use before any laboratory test would help decide who should be sent for testing.

Many machine learning studies report near-perfect CKD prediction using the UCI Chronic Kidney Disease dataset [3]. For example, Qin et al. reported 99.75% accuracy with Random Forest and 99.83% with a combined model [4]. However, the UCI features include serum creatinine, blood urea, urine albumin and specific gravity. These are the measurements used to diagnose CKD, so a model that uses them is largely re-deriving the diagnosis. At the point of screening, for a person who has never been tested, these values do not exist. Before starting this project, I ran the UCI dataset through a free no-code browser tool that trains a neural network, and it reached about 99% accuracy in under a minute. That experience, which is not part of the model comparison in this paper, is what led to the question of how much of that accuracy depends on diagnostic features.

Pre-laboratory risk scores already exist. SCORED, developed on NHANES data with logistic regression, uses age, sex, hypertension, diabetes, cardiovascular disease, anemia and proteinuria [5]. Anemia and proteinuria still require a test, however. This project asks two questions: (1) Can a model that uses only information available before any laboratory test identify adults with undiagnosed CKD well enough to guide referral for testing? (2) How much of the accuracy reported on the UCI dataset comes from features that are part of the diagnosis itself? The scope is limited to adults aged 20 and over who have never been told they have kidney disease, and to traditional machine learning algorithms.

II. METHODS

A. Datasets and ethics

The main dataset is the National Health and Nutrition Examination Survey (NHANES) 2017-March 2020 pre-pandemic release [6], a public, de-identified survey of the US population run by the National Center for Health Statistics, whose protocols are approved by its research ethics review board. I used 10 files (demographics, body measures, blood pressure, five questionnaires, serum biochemistry and urine albumin). The second dataset is UCI CKD [3]: 400 hospital patients with 24 features, released under CC BY 4.0. Both are public, and no identifiable information was used. All analysis rules (label, tiers, metric, split, models and selection rule) were written in a pre-registration document and committed to the project repository before any data were downloaded.

B. Cohort and label

From 15,560 NHANES participants, I kept adults aged 20 or older who were examined at the mobile examination center, not pregnant, had both serum creatinine and urine albumin-to-creatinine ratio (ACR), and had not been told they had weak or failing kidneys. This left 7,363 adults. CKD was defined as eGFR below 60 mL/min/1.73m2, calculated with the race-free CKD-EPI 2021 equation [7], or ACR of 30 mg/g or more. These laboratory values were used only to build the label and were never model inputs. The label is not circular: creatinine defines the answer, and the model learns which pre-laboratory characteristics predict that answer.

C. Feature tiers

Each feature was assigned to a tier before modeling. Tier 1 (definitional) features are KDIGO criteria or direct measures of kidney function. Tier 2 (consequential) features change because CKD is already present. Tier 3 (antecedent) features can be collected without a laboratory. Laboratory values that are neither were grouped as other laboratory. Table I lists the UCI assignment. The NHANES model uses 13 Tier 3 features: age, sex, BMI, waist, systolic and diastolic blood pressure, told of hypertension, blood pressure medication, told of high cholesterol, any cardiovascular disease, family history of diabetes, diabetes (no, borderline, yes) and smoking (never, former, current).

TABLE I
PRE-REGISTERED FEATURE TIERS FOR THE UCI DATASET

Tier	UCI features
1 Definitional	serum creatinine, blood urea, urine albumin, specific gravity, urine red cells
2 Consequential	hemoglobin, packed cell volume, red cell count, anemia, potassium, sodium, pedal edema, appetite
Other laboratory	urine sugar, blood glucose, white cell count, pus cells, pus cell clumps, bacteria
3 Antecedent	age, blood pressure, hypertension, diabetes, coronary artery disease

D. Exploratory data analysis and preprocessing

The EDA checked file sizes and duplicate IDs, answer codes for refused and don't know (set to missing), questionnaire skip patterns, impossible values, missing values, distributions, relationships with the label, correlations, outliers and subgroups. Data quality checks used fixed rules on the whole cohort; everything that looked at the label used the training set only. Preprocessing was the same for all models and was fitted inside a scikit-learn pipeline [8] on training data only: median imputation and standardization for numeric features, most-frequent imputation for yes/no features, and most-frequent imputation with one-hot encoding for categorical features.

E. Split, algorithms and tuning

Each dataset was split 80/20, stratified on the label, with random_state 42. The NHANES training set had 5,890 people and the test set 1,473. I compared Logistic Regression, Random Forest [9] and Histogram Gradient Boosting [10], all with balanced class weights to account for the 16.5% prevalence. Hyperparameters were tuned by grid search with the same 5 stratified folds for every model: C for Logistic Regression; maximum depth and minimum leaf size for Random Forest (300 trees); and learning rate, maximum depth and number of iterations for Gradient Boosting. Inside every fit, the decision threshold was tuned to maximize F2 with a separate inner cross-validation, so it was never chosen on the data used to score it.

F. Metrics and model selection

A false negative leaves CKD undetected, while a false positive costs one confirmatory blood and urine test. The primary metric was therefore F2, which weights recall more than precision but, unlike recall alone, still penalizes referring everyone. Supporting metrics were recall, precision, specificity, F1, accuracy, ROC-AUC, PR-AUC and the Brier score, reported as mean and standard deviation over the 5 folds. The selected model was the one with the highest mean F2; if the top two differed by less than one standard deviation, the simpler one was chosen (Logistic Regression, then Random Forest, then Gradient Boosting). Two baselines were scored on the same folds: referring everyone, and a risk-factor rule (refer if diabetes, told of hypertension, cardiovascular disease, or age 60 or older). The test set was used once, after selection.

G. Application

The selected pipeline, including preprocessing and the tuned threshold, was saved with joblib and loaded by a Streamlit [11] application. The application accepts one patient through a validated form or several through a CSV upload, and it rejects missing, invalid or out-of-range inputs with a readable message. It was tested with automated tests covering valid, invalid and boundary inputs.

III. RESULTS

A. Exploratory data analysis

Of the 7,363 adults, 1,217 (16.5%) had CKD: 766 by albuminuria only, 302 by low eGFR only and 149 by both. So most cases were early disease detected in the urine. CKD rose from 7% in adults in their 20s to 38% at 70 and over (Fig. 1). Diabetes was the strongest yes/no feature (CKD in 40% of people with diabetes against 12% without). No feature had a Spearman correlation with CKD above 0.28 (age), which is consistent with no feature leaking the label. BMI and waist

were highly correlated (0.91). Outliers flagged by the interquartile rule were all within plausible ranges and had higher CKD rates, so they were kept. People with a missing waist measurement had twice the CKD rate (31.6% against 16.0%), most likely because they could not be measured. I did not add missing-value indicators, because in the application a missing waist would only mean it was not measured. Fig. 2 shows that part of the age effect comes from the eGFR equation itself: a man with creatinine 1.3 mg/dL is labeled CKD from about age 67.

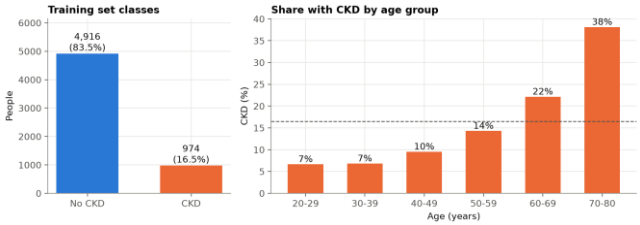


Fig. 1. Class balance in the NHANES training set (left) and CKD prevalence by age group (right). Dashed line: overall prevalence.

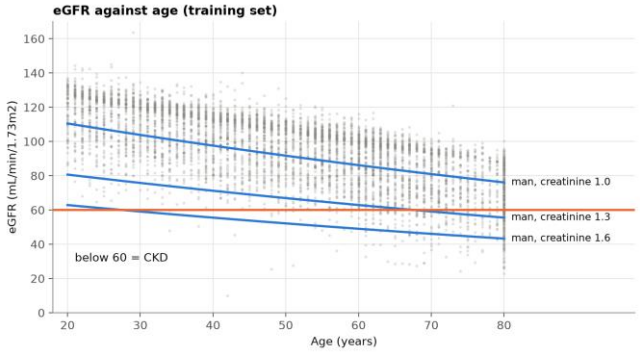


Fig. 2. eGFR against age in the training set. Curves show eGFR from the CKD-EPI 2021 equation for a man with fixed creatinine; the same creatinine crosses the CKD threshold at an older age.

B. UCI feature-tier ablation

With all 24 features, every model reached about 99% cross-validated accuracy and a ROC-AUC of 1.00, reproducing the literature (Table II, Fig. 3). Removing Tier 1 changed almost nothing, because hemoglobin alone had a ROC-AUC of 0.98. Removing Tier 1 and Tier 2 lowered accuracy to 0.74 to 0.94. With Tier 3 only, accuracy was 0.69 to 0.76, specificity 0.21 to 0.44, and F2 0.89 to 0.90, equal to referring everyone (0.893) at the UCI prevalence of 62.5%. On the UCI test set, the selected all-feature model reached 98.8% accuracy and the Tier 3 model 68.8%.

TABLE II
UCI ABLATION, 5-FOLD CROSS-VALIDATION MEANS

Feature set	Model	F2	ROC-AUC	Accuracy	Spec.
All features	LR	0.997	1.000	0.991	0.975
All features	RF	0.995	0.999	0.984	0.958
All features	GB	0.995	1.000	0.994	0.992
Without Tier 1	LR	0.984	0.995	0.969	0.933
Without Tier 1	RF	0.979	0.994	0.972	0.958
Without Tier 1	GB	0.981	0.992	0.969	0.942
Without Tier 1 and 2	LR	0.907	0.929	0.744	0.367
Without Tier 1 and 2	RF	0.950	0.980	0.906	0.808
Without Tier 1 and 2	GB	0.953	0.975	0.938	0.908
Tier 3 only	LR	0.898	0.897	0.691	0.208
Tier 3 only	RF	0.894	0.936	0.756	0.442
Tier 3 only	GB	0.902	0.922	0.738	0.358

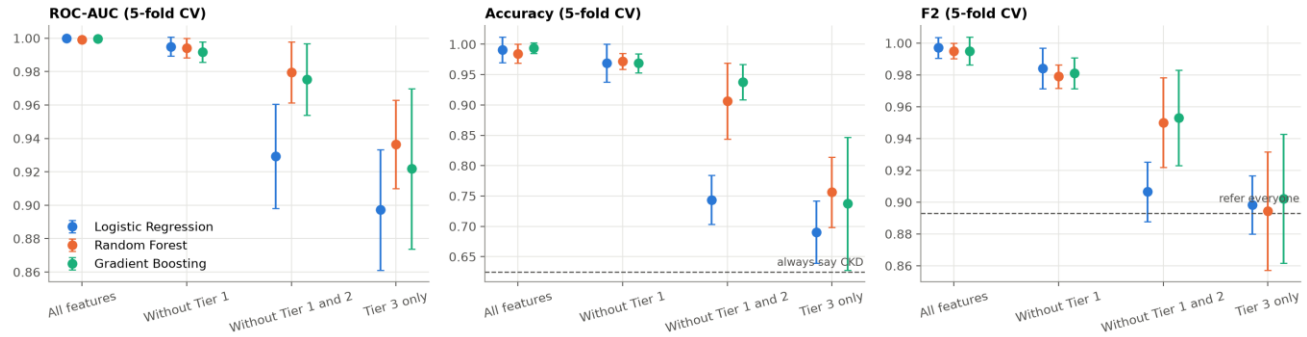


Fig. 3. UCI ablation: cross-validated ROC-AUC, accuracy and F2 (mean and one standard deviation) for each feature set and model.

C. NHANES model comparison

Table III and Fig. 4 show the cross-validation results. The three models were almost identical: mean F2 0.578 to 0.580 and ROC-AUC 0.758 to 0.770, with differences much smaller than the fold-to-fold spread. All beat referring everyone (0.498), but none clearly beat the risk-factor rule (0.576). Gradient Boosting had the highest mean F2 by 0.0001 over Random Forest, which was within one standard deviation, so the pre-registered rule selected Random Forest (maximum depth unlimited, minimum leaf size 20). Random Forest also had the highest ROC-AUC and PR-AUC, the lowest Brier score and the smallest spread across folds. All models were poorly calibrated because of the balanced class weights: a

score of about 0.8 corresponded to an observed CKD rate of about 0.5 (Fig. 5).

TABLE III
NHANES CROSS-VALIDATION, MEAN (SD) OVER 5 FOLDS

Model	F2	Recall	Specificity	ROC-AUC
Logistic Regression	0.578 (0.027)	0.793 (0.033)	0.591 (0.024)	0.758 (0.014)
Random Forest	0.580 (0.014)	0.819 (0.025)	0.554 (0.059)	0.770 (0.012)
Gradient Boosting	0.580 (0.023)	0.791 (0.053)	0.599 (0.036)	0.760 (0.014)
Risk-factor rule	0.576 (0.019)	0.841 (0.029)	0.514 (0.023)	-
Refer everyone	0.498 (0.001)	1.000 (0.000)	0.000 (0.000)	-

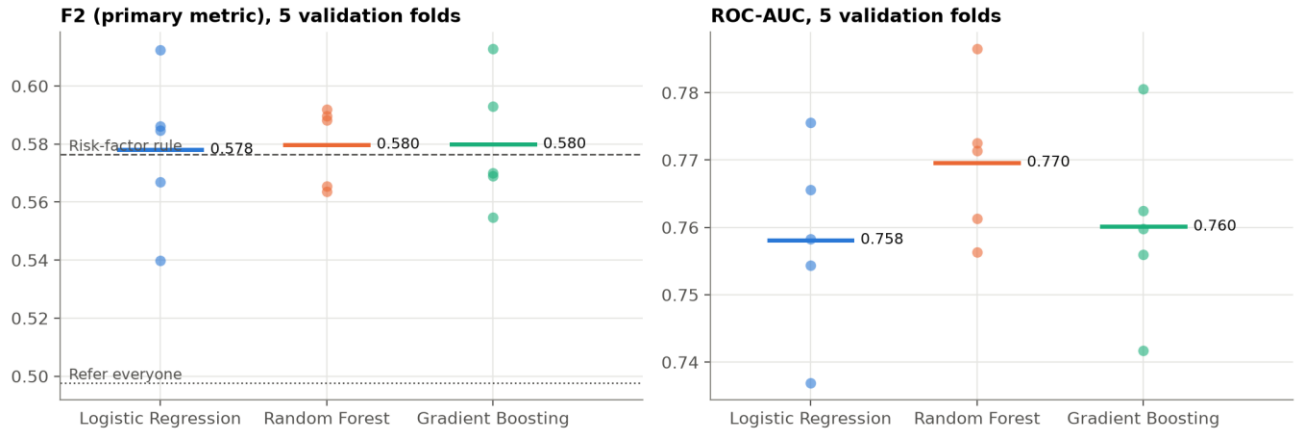


Fig. 4. F2 and ROC-AUC on each of the 5 validation folds. Lines show the means; dashed lines show the two baselines.

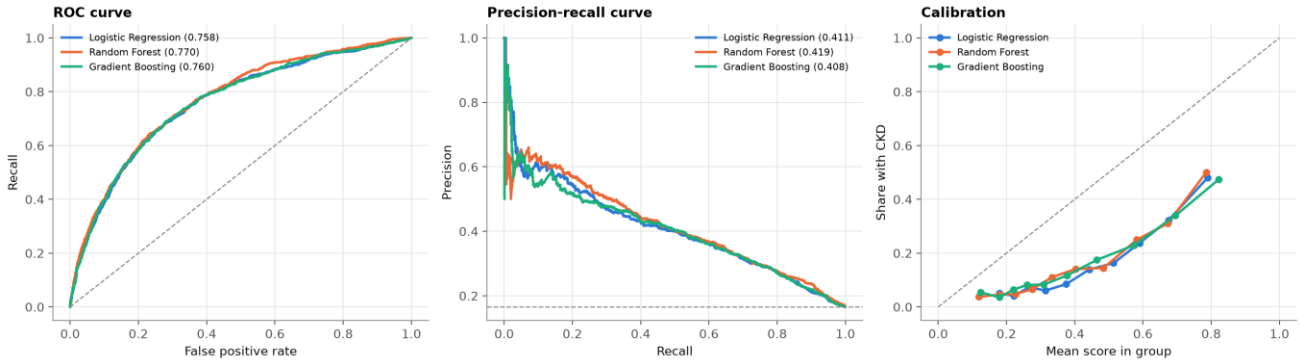


Fig. 5. Out-of-fold ROC, precision-recall and calibration curves for the three models.

D. Final test

On the 1,473 test participants, the Random Forest reached F2 0.586, recall 0.864, specificity 0.504, precision 0.256 and ROC-AUC 0.773, close to its cross-validation results. It caught 210 of 243 CKD cases and referred 820 people, 610 of whom did not have CKD (Fig. 6). The risk-factor rule reached F2 0.583 on the same people. In the 175 Non-Hispanic Asian test participants (26 cases), recall was 0.808 and ROC-AUC 0.795.

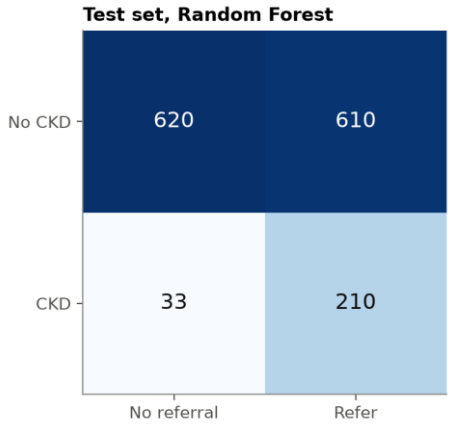


Fig. 6. Confusion matrix of the selected Random Forest on the test set at its tuned threshold (0.320).

E. Error analysis

In out-of-fold training predictions, recall was 0.95 for CKD defined by low eGFR only and 0.98 for both criteria, but 0.74 for albuminuria only. Recall was 0.97 at ages 60 to 80, 0.69 at 40 to 59 and 0.31 at 20 to 39. Permutation importance showed that the model relied most on age, then diabetes and systolic blood pressure (Fig. 7).

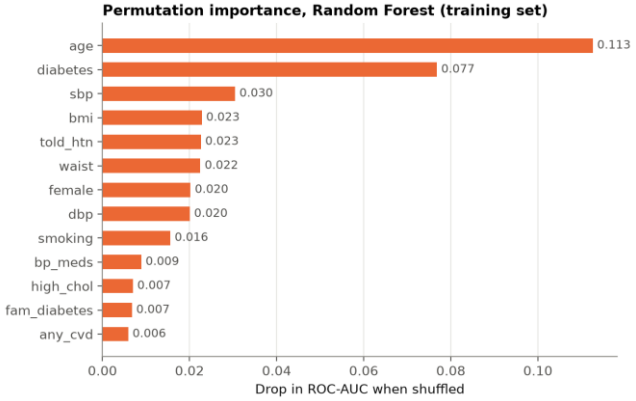


Fig. 7. Permutation importance of the Random Forest: drop in ROC-AUC when each feature is shuffled.

F. Application

The deployed application loads the saved pipeline and model card. Sample inputs and outputs, and screenshots of the application, are in Appendix B. The automated tests confirmed that the application loads, gives a recommendation for valid input, and shows an error instead of crashing for an out-of-range age or an empty form.

IV. DISCUSSION

The two datasets tell a consistent story. On UCI, the near-perfect accuracy depends on Tier 1 and Tier 2 features, which are measured after a patient is already suspected of kidney disease. Even the Tier 3 UCI models had ROC-AUC 0.90 to 0.94, much higher than on NHANES, because UCI compares hospital patients with clearly abnormal results against people without CKD, at a prevalence of 62.5%. A screening population is different: most people are healthy, and most CKD is early.

On NHANES, the three algorithms reached the same ceiling. When the features are a handful of risk factors, there is little non-linear structure for Random Forest or Gradient Boosting to exploit, and Logistic Regression already captures most of the signal. Random Forest was selected by the tie-break rule rather than because it was clearly better, and Logistic Regression would have been an equally defensible choice. The model also did not meaningfully beat the risk-factor rule on F2. Its practical advantage is a graded score: the tenth of training patients with the highest scores had CKD 50% of the time, against 3.7% in the lowest tenth, which lets a health unit with limited testing capacity test the highest-risk people first.

The errors follow the label. Low-eGFR CKD is concentrated in older adults, partly because the eGFR equation contains age, so age dominates the model. The model therefore misses most young adults with CKD and many people whose only sign is albuminuria. The tradeoff chosen by F2 is to refer about half of the people without CKD in order to catch 86% of CKD cases. Since the confirmatory test is cheap and a missed case can progress silently, I consider this acceptable for screening, but a negative result does not rule out CKD.

V. CONCLUSION AND RECOMMENDATIONS

Pre-laboratory screening for CKD is possible but limited. A Random Forest using 13 features that need no laboratory caught 86% of CKD cases in unseen data, with ROC-AUC 0.77, and was deployed as a working Streamlit application. It performs about as well as a simple risk-factor rule, and the near-perfect results reported on UCI depend on features that define or follow from the diagnosis. Limitations: the model is trained on US adults and not validated on Filipino patients; the label is based on a single measurement instead of the three-month requirement; survey weights were not used; and the model misses many young and albuminuria-only cases. Realistic improvements are external validation on Philippine data, adding a urine dipstick (cheap and available at many health units) as a low-cost laboratory step, recalibrating the scores, and using repeated measurements to define the label.

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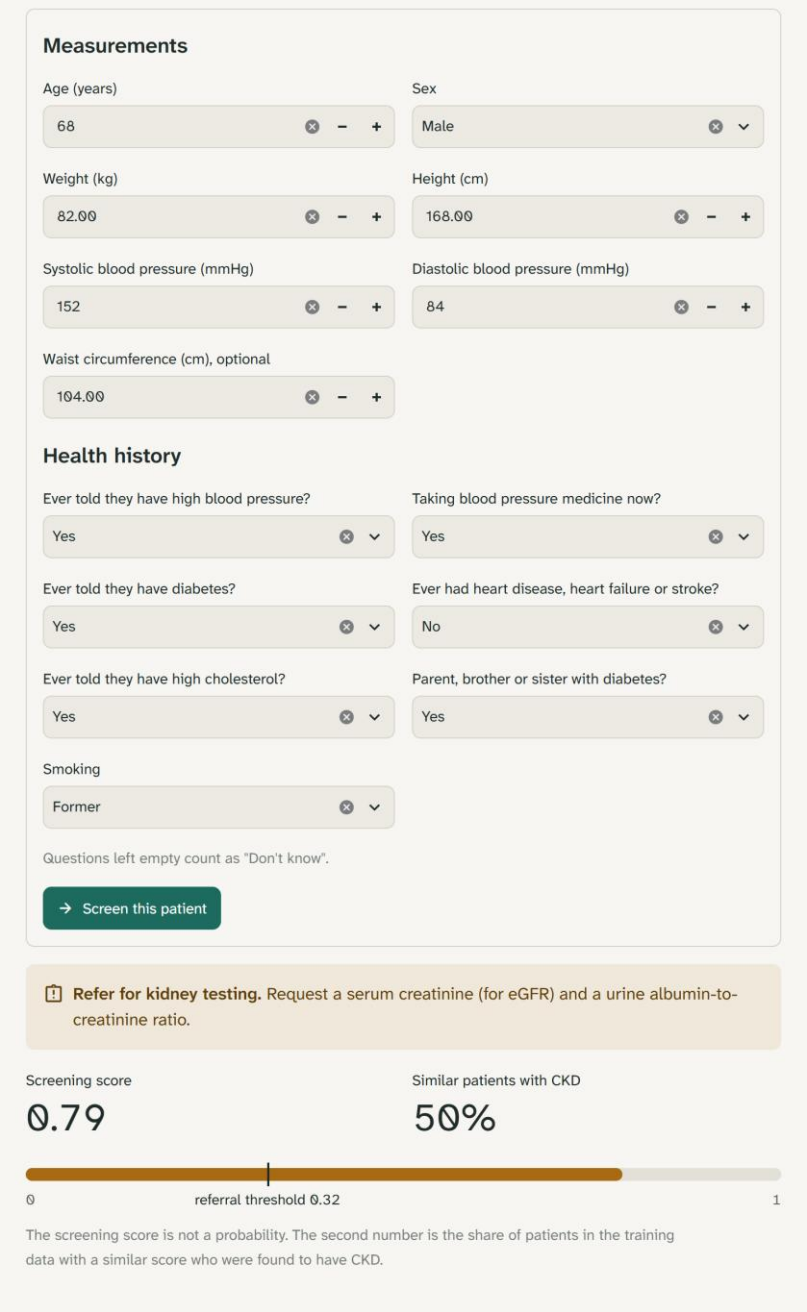
Appendix A. Data dictionary (NHANES model features)

Feature	Source	Unit or coding	Meaning
age	RIDAGEYR	years	Age at screening. 80 means 80 or older.
bmi	BMXBMI	kg/m2	Body mass index
waist	BMXWAIST	cm	Waist circumference
sbp	BPXOSY1-3	mmHg	Systolic blood pressure, mean of up to 3 readings
dbp	BPXODI1-3	mmHg	Diastolic blood pressure, mean of up to 3 readings
female	RIAGENDR	1 = female, 0 = male	Sex
told_htn	BPQ020	1 = yes, 0 = no	Ever told by a doctor they have high blood pressure
bp_meds	BPQ040A, BPQ050A	1 = yes, 0 = no	Currently taking prescribed blood pressure medicine
high_chol	BPQ080	1 = yes, 0 = no	Ever told by a doctor they have high cholesterol
any_cvd	MCQ160B-F	1 = yes, 0 = no	Ever told they had heart failure, coronary heart disease, angina, heart attack or stroke
fam_diabetes	MCQ300C	1 = yes, 0 = no	A close relative has diabetes
diabetes	DIQ010	no / borderline / yes	Ever told by a doctor they have diabetes
smoking	SMQ020, SMQ040	never / former / current	Never = under 100 cigarettes in their life

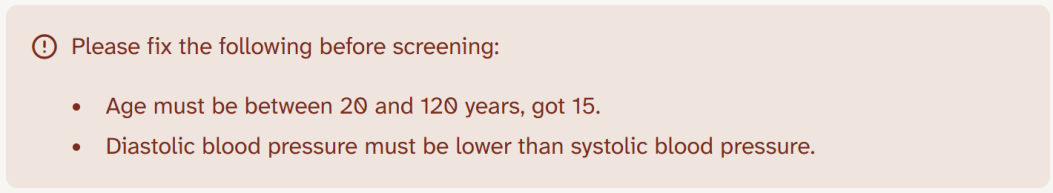
Full data dictionary and processing notes: documentation/data_dictionary.md.

Appendix B. Sample inputs and outputs

Input	Output
age 68, male, 82 kg, 168 cm, waist 104, BP 152/84, told high BP yes, BP medicine yes, high cholesterol yes, heart disease no, family diabetes yes, diabetes yes, smoking former	Refer for kidney testing; score 0.79 (threshold 0.32); similar patients with CKD 50%
age 45, female, 70 kg, 155 cm, waist 92, BP 132/82, told high BP no, BP medicine no, high cholesterol no, heart disease no, family diabetes dont_know, diabetes borderline, smoking never	Refer for kidney testing; score 0.49 (threshold 0.32); similar patients with CKD 14%
age 28, male, 65 kg, 172 cm, waist not entered, BP 116/72, told high BP no, BP medicine no, high cholesterol no, heart disease no, family diabetes no, diabetes no, smoking current	No referral indicated; score 0.13 (threshold 0.32); similar patients with CKD 4%. Note: Waist was not entered, so the typical value from the training data is used.
age 91, female, 50 kg, 150 cm, waist 80, BP 160/70, told high BP yes, BP medicine yes, high cholesterol no, heart disease yes, family diabetes no, diabetes no, smoking never	Refer for kidney testing; score 0.79 (threshold 0.32); similar patients with CKD 50%. Note: Age 91 is treated as 80, the oldest age in the training data.
age 15, female, 50 kg, 150 cm, waist 70, BP 110/120, told high BP no, BP medicine no, high cholesterol no, heart disease no, family diabetes no, diabetes no, smoking never	Rejected: Age must be between 20 and 120 years, got 15. Diastolic blood pressure must be lower than systolic blood pressure.



Screenshot of the application for the first sample patient: the form, the referral recommendation, the screening score against the threshold, and the share of similar training patients with CKD.

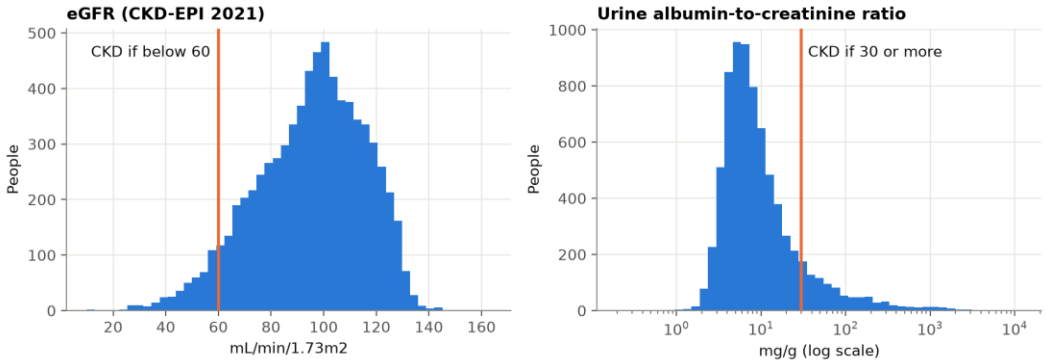


The fifth sample patient entered in the form. The inputs are rejected with a readable message instead of a prediction.

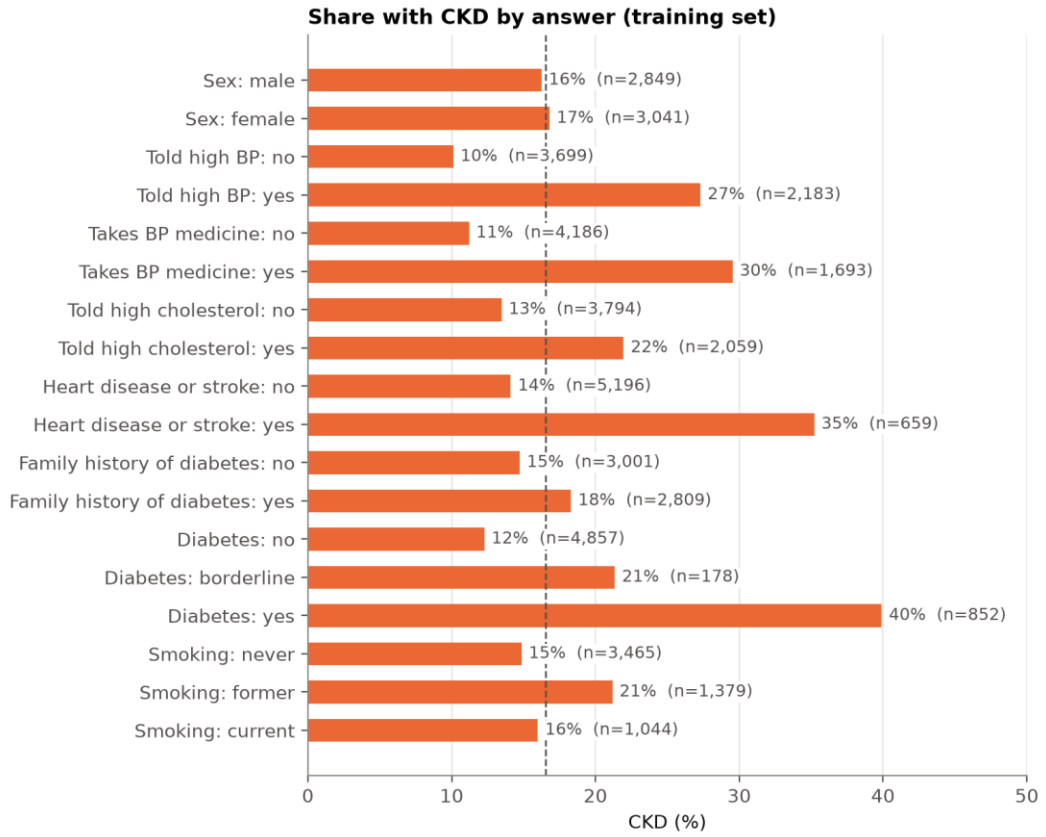
	Recommendation	Score	Similar patients with CKD	age	sex	weight_kg	hei
1	refer for kidney testing	0.79	50%	68	male	82	
2	refer for kidney testing	0.49	14%	45	female	70	
3	no referral	0.13	4%	28	male	65	
4	refer for kidney testing	0.79	50%	91	female	50	
5	skipped, see the message above	None	None	15	female	50	

Result columns after uploading the five sample patients as a CSV file. The four valid rows are screened and row 5 is skipped with the same message.

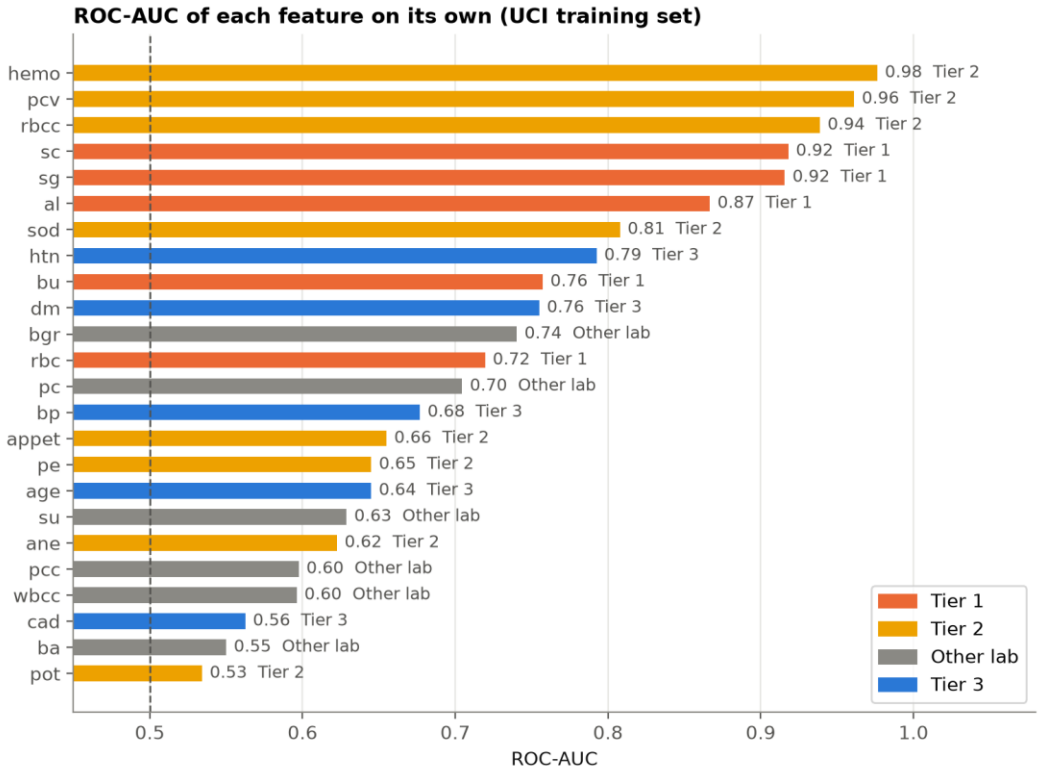
Appendix C. Additional figures



Distributions of eGFR and urine ACR in the cohort, with CKD thresholds.



Share with CKD by answer to each yes/no and categorical feature (training set).



ROC-AUC of each UCI feature on its own, colored by tier.

Appendix D. Contribution record

Member	Assigned responsibilities	Actual contributions	Confirmation
Antonio Garcia	All project work (sole member)	Problem definition and pre-registration; data acquisition; EDA; preprocessing; model training, comparison and selection; final evaluation; Streamlit application; tests; technical documentation; paper	Signature: _____

Ownership and Authorship Declaration

Project Information	Details
Project title	Pre-lab CKD Early Detection using Random Forest
Course and section	_____
Group name or number	_____
Date submitted	_____

Declaration Statement

We, Dr. Armida P. Salazar as the course instructor and the undersigned student group members, jointly declare that we are the owners and authors of the submitted project output identified above. The project output includes the application, source code, trained model and saved pipeline, exploratory data analysis, visualizations, technical documentation, research paper, and original materials produced specifically for this course project.

We affirm that the submitted work accurately represents our documented contributions and that all external datasets, software libraries, code, models, frameworks, images, publications, and other third party materials have been properly cited, licensed, or acknowledged using IEEE citation format where applicable. This declaration does not transfer or claim ownership of pre-existing third party or institutional materials and remains subject to applicable National University policies, intellectual property rules, licenses, and Philippine law.

Signatories

Name and Role	Signature	Date
Dr. Armida P. Salazar Course Instructor		
Antonio Garcia Student Group Member		